Change Gear Chart

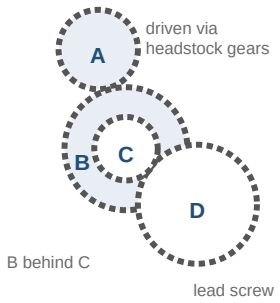
5100 HiTorque Mini Lathe
16 TPI lead screw

Unified Inch Threads (UN)

TPI	A	B	C	D
4	80	any	any	20
4.5	80	40	80	45
5	80	20	40	50
6	80	20	40	60
7	80	any	any	35
8	40	any	any	20
9	80	any	any	45
10	80	any	any	50
11	80	any	any	55
11.5*	45	35	65	60
12	60	any	any	45
13	80	any	any	65
14	40	any	any	35
16	40	any	any	40
18	40	any	any	45
20	40	any	any	50
24	65	60	40	65
27	40	45	40	60
28	20	any	any	35
32	20	any	any	40
36	20	any	any	45
40	20	any	any	50
44	20	any	any	55
48	40	60	40	80
56	40	35	20	80
64	20	any	any	80
72	40	45	20	80
80	20	80	40	50

GEAR POSITIONS

- **A** is the first (top) change gear. It is driven from the spindle through internal headstock gears, not mounted on the spindle.
- **B** (inside) and **C** (outside) share the middle shaft. A drives B, and C drives D.
- **D** sits on the end of the lead screw.
- **any**: three-gear setup. Any gear in B meshes A with D without changing the ratio, and any gear in C is only a spacer. Move the D spacer outside the gear so D lines up with B.



Metric Threads (pitch, mm)

PITCH	A	B	C	D	ACTUAL	DEVIANCE
0.3	20	57	35	65	0.2999	−0.02%
0.35	20	80	50	57	0.3481	−0.53%
0.4	20	55	45	65	0.3997	−0.09%
0.45	20	60	55	65	0.4478	−0.50%
0.5	21	50	45	60	0.5001	+0.01%
0.6	21	50	45	50	0.6001	+0.01%
0.7	35	55	45	65	0.6994	−0.09%
0.75	35	65	50	57	0.7498	−0.02%
0.8	40	65	45	55	0.7993	−0.09%
1	21	50	60	40	1.0001	+0.01%
1.25	35	40	45	50	1.2502	+0.01%
1.5	40	50	65	55	1.5009	+0.06%
1.75	40	35	55	57	1.7506	+0.04%
2	40	50	55	35	1.9957	−0.21%
2.5	60	55	65	45	2.5015	+0.06%
3	65	55	80	50	3.0018	+0.06%
3.5	65	21	57	80	3.5010	+0.03%
4	55	50	80	35	3.9914	−0.21%
4.5	55	21	65	60	4.5042	+0.09%
5	55	35	80	40	4.9893	−0.21%
5.5	50	55	80	21	5.4978	−0.04%
6	55	21	65	45	6.0056	+0.09%
8	57	21	65	35	8.0023	+0.03%

← *11.5 TPI setup actually cuts 11.487 TPI (−0.11%).

TURNING FEEDS (POWER FEED)

	A	B	C	D	in/rev
Normal	20	80	20	80	0.0039″
Fine	20	80	15	90	0.0026″

Normal is the factory setup. Put it back after threading. Fine gives a better finish and needs the LittleMachineShop #1137 set: an 80/15 compound gear for B–C and a 90 for D.

GEAR SET

20 (2), 21, 30, 35, 40 (2), 45, 50, 55, 57, 60, 65, 80 (2). The 20s and 80s come installed.

RATIOS

Four gears: `TPI = 16 × (B × D) / (A × C)`
Three gears: `TPI = 16 × D / A`
Feed per rev = 1 / TPI. Pitch (mm) = 25.4 / TPI, so metric pitches are approximate. *Actual* is what each setup cuts, and *Deviance* is how far that is from the nominal pitch.